

HMRPred: User Tutorial for Predicting Heavy Metal Resistance Proteins

Tutorial

HMRPred is a machine learning-based web resource designed for the identification of Heavy Metal Resistance (HMR) proteins across ten heavy metals: Arsenic, Cadmium, Chromium, Copper, Iron, Lead, Mercury, Nickel, Silver, and Zinc. The platform supports both FASTA sequence input and file upload, enabling genome-wide scans and single-sequence prediction.

Accessing HMRPred

Visit the HMRPred web server: <https://hmrpred.streamlit.app/>

The homepage provides an overview of the tool, including supported metals, features, and applications in **bioremediation, phytoremediation, and synthetic biology**.

Navigation Menu

On the left panel, you will find:

- **Home** – Introduction & overview
- **Prediction** – Run HMR protein predictions
- **Developers** – Developer details
- **Contact Us** – Communication details

Click on **Prediction** to begin the analysis.



Figure 1: Homepage.

Step-by-Step Usage Guide

Step 1: Select Metal Type

Choose the heavy metal of interest from the dropdown list.

Example shown in screenshot: **Zinc**

This selection loads the model trained specifically for that metal.



Figure 2; Select the metal of interest.

Step 2: Input Protein Sequence(s)

HMRPred supports two input modes:

Option A: Paste FASTA Sequence(s)

- Select **Paste FASTA Sequence**
- Enter one or multiple protein sequences in FASTA format:

```
>PROT1
MKKLIYQTNQFANGELQVENTSYELCDTFKEL...
>PROT2
MTKITAKVTKSDKQWNWGD TTFWVDGNETGF...
```

Tip: Ensure each sequence starts with > followed by a protein name.



Figure 3: Load the input sequence.

Option B: Upload File

- Select **Upload FASTA File**
- Upload .fasta or .fa file
(Up to **200 MB** supported)

Step 3: Run Prediction

Click **Predict**

HMRPred will analyze the input and classify each sequence as **HMR protein** or **non-HMR protein**, using metal-specific ML models.

Step 4: View Results

The output table displays:

Column	Description
Sr. No.	Sequence index
FASTA Sequence	Input protein sequence
Probability (No HMR)	Probability of negative class
Probability (HMR)	Probability of positive class
Prediction	Yes / No HMR protein

Example Output

Sample prediction results:

Sequence	Prob(Non-HMR)	Prob(HMR)	Prediction
PROT1	0.445	0.555	Yes
PROT2	0.550	0.450	No
PROT3	0.710	0.290	No
PROT4	0.565	0.434	No
PROT5	0.410	0.590	Yes

Higher Prob(HMR) indicates stronger confidence of metal resistance.

